

Sire3 Vineyard AI Field Challenge metrics report

Team Solemtrix · GigaHack 2026 · 27 September 2026, 14:00

Numbers are grouped by the six criteria of the organisers’ “Scoring weights” slide. **Official metrics** are computed with the challenge metric definitions on the organisers’ 2 reference tiles (vineyard eval-examples -gates). **Survey numbers** cover all 311 tiles (81.5 ha) and come from the published deliverables in the repository root: measurements.csv, route.geojson, targets.geojson (annotation run rc10f, route run 20260927T1400-post-rc10f-pretest, route from the organizers’ pre-test START). All geometry is EPSG:32635, measured planar.

Canopies 25%	Waste 10%	Axes 15%	Measures 10%	Route 25%	Engineering 15%
85% objective output metrics				15% expert engineering assessment	

At a glance

Criterion	Weight	Headline result
Canopies	25%	official canopy score 0.876 (IoU 0.856, F1 0.905); inter-row IoU 0.960; 13,702 plants on the survey
Waste	10%	1,331 candidates → CLIP probe → SAM 3 → team approval: 11 boxes exported, 0 unreviewed
Axes	15%	row F1 1.00 (51/51), grouping F1 1.00, attributes 1.00; 674 rows in 44 blocks on the survey
Measures	10%	official counts score 0.973 ; row-length error 0.05%, canopy-area error 1.4% (0.3% pooled)
Route	25%	24.18 km closed route from the pre-test START, 795 targets visited, 1.29% outside (limit 2%), 65% shorter than the baseline tour
Engineering	15%	tiles → deliverables in ≈20 min on a laptop; 2,378 tests, CI, export validator (311 tiles, 0 errors)

Canopies25%

0.876	0.856	0.905	0.960	585/650
canopy score (official)	canopy IoU	canopy F1 (0.945 on r006_c004)	inter-row IoU	reference plants matched

- **Method:** Lab a* vegetation threshold restricted to each row corridor; connected components give one polygon per plant. No GPU needed.
- **Inter-rows:** band between adjacent axes, inset 0.30 m, forbidden zones notched out F1 **1.00** (49/49 matched).
- **Model choice backed by an ablation** (mean canopy score on both reference tiles): classical CV 0.816, CV ∪ U-Net 0.773, CV ∩ U-Net 0.707, U-Net ∩ corridor 0.675, U-Net alone 0.641. The U-Net (vine-unet v1) is used only as row evidence in cadastral completion regions.
- **Survey:** 13,702 plant polygons, canopy area (union) 15,427.54 mš (1.5428 ha), inter-row area 93,244.01 mš (9.3244 ha).

Waste10%

Stage	Crops	Role
Rule candidates (colour, shape, off-lattice)	1,331	high-recall first pass over all tiles
OpenCLIP ViT-B-32 + logistic probe	ranked	trained on 11,292 crops (3,000 UAVWaste positives, 8,292 Sire3 negatives)
SAM 3 verification	top crops	13.1 s per 512px crop on CPU, so only the best-ranked ones
Team approval		every box is checked before export (waste_confirmed.csv)
Exported to Marcaj	11	precision over recall: no unreviewed box reaches the scorer

Waste inside vineyards counts from 10×15px (stones and clods included, by team rule). The 11 waste boxes are exported to Marcaj and shown on the map; the route does not visit them (team rule), so they are not in targets.geojson.

Row axes15%

1.00	1.00	51/51	49/49	0.05%
row F1 (51/51 rows)	grouping F1 (blocks 2/2)	row structure correct	inter-row cover correct	row-length error

- **Detection:** coarse-to-fine dominant angle (2ř, then 0.5ř), 1-D profile on the row normal, autocorrelation spacing (1.8–3.8 m), SNR ≥ 3 rejects orchards, one Huber line per peak.
- **Linking:** union-find in UTM joins rows across tile edges, so each row keeps one vineyard_id/row_id; rows are cut where a road crosses them (a road always separates blocks).
- **Propagation:** tiles without a confident row are searched again at the neighbours’ lattice (angle ±2ř, spacing ±15%); AGCC cadastral parcels with ≥ 3 rows seed the completion of their uncovered part.
- **Survey:** 674 rows, 44 blocks (V01–V44), 47,737.51 m of axes; 124 of the 311 tiles hold vine rows.

Measurements

10%

0.973

counts score (official)

0.05%

row-length error

1.4%

canopy-area error (0.3% pooled)

0.5%

inter-row-area error

51/51

rows counted

measurements.csv (survey → block → row, joined by vineyard_id/row_id); canopy area is the union of polygons, so tile overlaps count once.

Level / ID	Rows	Row length	Canopy	Inter-row	Plants
survey (44 blocks)	674	47,737.51 m	15,427.54 mš	93,244.01 mš	13,702
block V01	115	9,955.4 m	3,262.9 mš		2,346
block V13	77	13,551.2 m	3,471.8 mš		5,525
block V41	83	4,089.9 m	1,469.4 mš		831

Route

25%

24.18 km

closed route length

795

targets visited

1.29%

outside allowed area (limit 2%)

0.00 m

closure at START

–65%

vs baseline tour

Target type (targets.geojson)	Total	Visited	Share
missing plant	929	628	68%
row gap	228	141	62%
row end short	51	22	43%
missing row	7	4	57%
all	1,215	795	65.4%

- **Walking domain:** (inter-rows $\cup$ authorised passages) – forbidden zones – canopies; targets solved as a generalised TSP with OR-Tools (python-tsp fallback).
- **Length:** 24,184.92 m (362.8 min at 4 km/h) against 68,174.61 m for the pipeline baseline tour (–65%) and 62,234.20 m for visiting the targets in ID order (–61%). The 11 waste boxes are reported but not routed.
- **Validity gates in publish:** single LineString, EPSG:32635, start and end exactly at the organizers' pre-test START (629663.80, 5220195.30), outside share $\leq 1.5\%$ (stricter than the 2% rule). Measured: 1.29% outside, 0.00 m closure.

Engineering

15%

Stage (same laptop, full 311 tiles)	Time	Notes
U-Net inference (88 completion tiles)	12.9 s	MPS/CPU, optional extra
Perception (vineyard run)	384 s	rows → blocks → canopy → inter-row → attributes → waste, 8 workers
CVAT / Marcaj export	24 s	validator: 311 tiles, 0 errors
Post chain, route from the pre-test START	806 s	derive → passable → targets → route (574 s, final 60 s solver) → measure → farms → web
Tiles → route.geojson + measurements.csv	≈20 min	timings from logs/pipeline.jsonl

- **Architecture:** typed stage contracts, one CLI command per stage, torch-free core; ML (U-Net, OpenCLIP, SAM3) and OR-Tools are optional extras.
- **Robustness:** 2,378 automated tests, CI on every push, export validator and publish gates (closure, CRS, outside share).
- **Scalability and performance:** linear in area, per-tile workers, caches keyed by config hash; the route stage dominates and is time-boxed (30 s per OR-Tools solve, 60 s in final mode).
- **Reproducibility:** pinned uv.lock, Docker image runs with `--network none`, open weights (vine-unet v1 as a GitHub release, sha256-verified), MIT licence.
- **Web app** (Next.js, RO/EN/RU): every layer, the official route with GPX/GeoJSON download, farms, roads and the live AGCC cadastre.